SRM TRP Engineering College
Tiruchirappalli
Course Assignment Instructions with Rubrics

Department	Artificial Intelligence and Data Science.										
Programme:	BTech		**Semester:**	V	**AY:**						
Faculty Handling:	Mrs. R. Kalaiyarasi										
Course Code & Title:	AD3501 - Deep Learning										
Assessment Task:			**Assessment Type:**	Term Work/Self			**Required Hrs:**				
Assessment Question based on:	*Open Ended*	✓	*Problem Based Learning*		✓	*Creative Thinking*					
	Close Ended		*Project Based Learning*			*Critical Thinking*			✓		
	Any Other Type	MCQ or any other	*Challenge Based Learning*		✓	*Level of Thinking (RBT)*			K5		
SDG to be Addressed:	SDG is Required	Yes/No	*SDG Number (Specify or Mention Students Choice)*			*Target Number (If Applicable)*					
SDG Integration Level: (If Addressed)	*Level 1: Awareness*		*Level 2: Design Alignment*		✓	*Level 3: Impact and Measurement*					
RIASEC Context	Realistic	Investigative ✓	Artistic		Social		Enterprising		Conventional ✓		
Presentation Format:	PowerPoint/Poster/Video/PDF/Written/Any other (Specify)										
Max Marks:	~~40~~										
Relevant Course Outcome(s):											

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Submission Deadline:	

Instructions to the Students:
Assessment Question: Automatic answer sheet Evaluation using Deep learning

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Guidelines:

E.g. Guidelines for Preparation/Submission/Expected Output/Review Process/Presentation Format if any

Rubrics for Evaluation:

Performance Criteria	Excellent (5)	Good (4)	Satisfactory (3)	Needs Improvement (2)	Unsatisfactory (1)

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Literature Survey Table

S.No	Paper Title & Year	Author(s)	Journal / Quartile	Methodology	Dataset(s)	Model	Result	Limitations
1	Artificial intelligence assisted automated short answer question scoring tool shows high correlation with human examiner markings (2025) — Base Paper	H. M. T. W. Senevirathne, S. S. Manathunga	BMC Medical Education (Q1)	The authors set the LLM up as an expert examiner by putting the question, rubric, and model answer into the system prompt, then ran a 3-stage prompt chain (extract relevant excerpts → assign rubric marks → give feedback), sampling each answer 5 times and taking the median score.	30 pharmacology short-answer scripts (60 individual answers) from University of Peradeniya, marked against instructor-written rubrics.	GPT-4, with temperature sampling (mean 0.3, SD 0.1) across 5 runs.	Pearson correlation of 0.93 and 0.96 against two independent human examiners; ICC = 0.94, described as excellent inter-rater reliability; Kruskal-Wallis found no significant median difference (p = 0.17).	Only 30 scripts from one course at one institution; a measurable scoring bias turned up against one of the two human raters that the authors could not fully explain; performance is expected to weaken for topics the LLM has little training exposure to.
2	Cross-prompt Pre-finetuning of Language Models for Short Answer Scoring (2025)	H. Funayama, Y. Matsubayashi, Y. Asazuma, T. Mizumoto, K. Inui	International Journal of Artificial Intelligence in Education (Q1)	I read this as a two-phase training strategy: first pre-finetune a BERT or LLM on answers from many other exam prompts using key phrases pulled from each rubric, then finetune that pre-trained model on only a small number of examples from the actual target prompt.	RIKEN Japanese short-answer dataset — 109 analytic-criterion prompts, expanded with 10,000 new annotations by the authors.	BERT-based regression model; Llama-3-Swallow 8B and 70B LLMs with QLoRA finetuning.	The 70B model reached near-human QWK using roughly 100 labelled answers for the new prompt, about half the data a standard finetuned model would need for the same accuracy.	Everything was tested on one language (Japanese) and one domain (reading comprehension); it is not shown whether the same gains would carry over to a technical subject or a different language.
3	SAS-Bench: A Fine-Grained Benchmark for Evaluating Short Answer Scoring with Large Language Models (2025)	P. Lai, K. Zhang, Y. Lin, L. Zhang, F. Ye, J. Yan, Y. Xu, C. He, Y. Wang, W. Zhang, B. Cui	arXiv preprint / Peking University, Fuzhou University (benchmark, not yet journal-indexed)	Rather than building a new scorer, the authors built a benchmark: real exam questions annotated step-by-step with both a score and a labelled error cause per step, then tested 16 different LLMs against two new metrics that look at step-wise consistency, not just the final overall score.	4,109 expert-annotated student responses across 9 subjects, drawn from China's Gaokao (national entrance exam) question bank.	16 LLMs compared, including GPT-4o, DeepSeek-R1, DeepSeek-V3, Qwen3, and Llama3.3-70B.	Models were noticeably weaker at scoring science subjects that need multi-step reasoning than humanities subjects, and every model's step-wise consistency score was lower than its overall-score accuracy — a gap that a simple accuracy number would hide.	As a benchmark rather than a deployable system, it does not propose a scoring pipeline of its own; it is limited to the Chinese Gaokao question style, so its findings may not transfer directly to open-ended university-level answers.
4	AI-Driven Subjective Answer Evaluation: Integrating OCR and Semantic Similarity for Intelligent Learning Environments (2026)	A. Yadav, Shalu, S. Rathee	Proceedings on Engineering Sciences (Scopus-indexed)	The authors used OCR to convert both the teacher's answer key and the student's handwritten script into text, generated SRoBERTa sentence embeddings for both, and scored the answer using cosine similarity between the two embeddings, then converted that into a percentile-based grade.	150 handwritten answer sheets from 10 undergraduate courses across STEM and humanities subjects, with 1,800 question-level ratings from teaching assistants as ground truth.	SRoBERTa sentence embeddings + cosine similarity (compared against Euclidean and Manhattan distance).	Cosine similarity gave the best result of the three metrics (0.83 Pearson vs 0.76 and 0.72), and the full pipeline reached 0.89 Pearson correlation with expert graders overall.	A controlled experiment showed accuracy dropping steadily as OCR error rate rose, from 0.89 correlation on clean text down to 0.81 with a weaker OCR engine — so the whole system's ceiling is set by OCR quality.
5	Automated Evaluation of Handwritten Answer Script Using Deep Learning Approach (2024)	M. G. Siddaraj, P. Sukesh, B. K. Nithin, M. S. Shreyas, A. V. Hadapad	International Research Journal of Modernization in Engineering Technology and Science (non-indexed)	This paper describes a CNN + RNN pipeline: handwriting is optically recognised and converted to digital text, then the extracted answer is compared against a stored key answer and marked based on the percentage of the key answer matched.	Not specified in the paper — no dataset size, source, or subject area is given.	CNN for visual feature extraction paired with an RNN (BiLSTM-style) for sequence-to-text conversion.	The paper describes a working end-to-end architecture and data-flow design, but does not report any accuracy, correlation, or agreement figures against human graders.	Without any quantitative validation, it is impossible to judge how reliable the system actually is; it is included here mainly to show the range of maturity across the literature, from validated studies to concept-stage pipelines.
6	Towards the Implementation of Automated Scoring in International Large-Scale Assessments: Scalability and Quality Control (2025)	J. Y. Jung, L. Tyack, M. von Davier	Computers and Education: Artificial Intelligence, Elsevier (Scopus Q1)	The authors combined machine translation (choosing Google Translate or ChatGPT per language, based on whichever agreed more with human scorers) with a bag-of-words neural network classifier, and separately tried an unsupervised LLM-only approach with fuzzy matching to group misspelled variants of the same answer.	Three constructed-response science items from TIMSS 2023, answered in 42 languages across more than 50 participating countries.	Feed-forward neural network (scikit-learn) on machine-translated bag-of-words features; separate ChatGPT-based unsupervised scorer.	The combined-translation approach reached 93.85% exact agreement and a Cohen's kappa of 0.86 against human raters, the best of the four methods compared.	Machine translation hallucinated for several low-resource languages, and a meaningful share of AI-human disagreement traced back to human raters disagreeing with each other across countries, not machine error — a scale of study none of the smaller papers here can replicate.
7	Can AI Grade Like a Professor? Comparing Artificial Intelligence and Faculty Scoring of Medical Student Short-Answer Clinical Reasoning Exams (2025)	A. Rajan, S. M. Alexander, C. L. Shenvi	Advances in Health Sciences Education (Q1)	GPT-4o was used to directly grade narrative short-answer exam responses without any fine-tuning, and its scores were checked against faculty scores using a pre-registered statistical equivalence test rather than a simple correlation.	1,450 individual short-answer responses from case-based learning clinical-reasoning exams.	GPT-4o (zero-shot grading).	Overall, GPT-4o's scores were statistically equivalent to faculty scores (mean difference −0.55%, 95% CI within ±5%), with high precision (ICC = 0.993); equivalence held for lower Bloom's-taxonomy levels but broke down for	Agreement with faculty got noticeably worse as question difficulty increased, and the authors are explicit that this supports using AI as a preliminary-scoring aid rather than a stand-alone replacement for faculty grading.

S.No	Paper Title & Year	Authors	Journal / Source	Methodology	Datasets	Model	Result	Limitation
							"Understanding" and "Evaluating" questions.	
8	*RUBRA: An Agentic AI System for Automatic Short Answer Grading Using LLMs and RAG (2025)*	S. Halder, S. Sengupta, A. K. Das	*2025 Conference on Intelligent Computing and Applications*, IEEE (Scopus-indexed)	Instead of relying only on what the LLM already knows, RUBRA retrieves relevant passages from textbooks and course material through a RAG pipeline and dynamically generates a rubric from the tutor's reference answer, then aligns the student's response against that rubric	SCIENTSBANK benchmark dataset for automatic short-answer grading	LLM combined with a retrieval-augmented generation (RAG) pipeline and dynamic rubric generation.	*The retrieval-grounded, dynamically-rubricked design outperformed prior AERG approaches on prediction accuracy while also making the generated feedback more explainable.*	*Having several moving parts — retrieval, dynamic rubric generation, scoring — makes the system more complex to deploy and audit than a single-prompt approach, and its extra computation cost is not directly compared against simpler baselines.*
9	Fine-Grained Rubric-Aware Automated Scoring of Programming Short Answer Questions via Large Language Models in Vocational Education (2026)	C. Zhang, L. Sun	*ACM* (conference/publication record)	The authors broke a standard answer down into what they call Atomic Scoring Units, then used a reasoning-alignment module to map pieces of the student's own response onto those individual units, rather than judging the whole answer in one pass.	Programming short-answer questions spanning five knowledge dimensions, in a vocational-education setting.	Fine-grained Adaptive Scoring Framework (GASF), an LLM-based scoring pipeline with an atomic-unit decomposition step.	GASF reached a peak Quadratic Weighted Kappa of 0.7910 and cut mean absolute error by up to 78.9% on complex logical questions compared with a zero-shot LLM baseline.	The atomic-unit decomposition is built specifically around how programming answers are structured, so it is not obvious the same technique would transfer cleanly to free-text descriptive subjects outside programming.
10	Are Rubrics All You Need? Towards Rubric-Based Automatic Short Answer Scoring via Guided Rubric–Answer Alignment (2026)	S. Gombert, Z. Sun, F. Zehner, J. Lonsjoe, T. Wyrwich, B. Celincel, D. Bednorz, M. Rathsack, D. Di Mitri, K. Neumann, H. Drachsler	*ACM* (conference/publication record)	The authors argue that most prior work treats the rubric as a side input rather than the main scoring reference, so they built two transformer architectures (*GRAASP* and *ToLeGRAA*) that use attention specifically to align a student's answer against individual rubric criteria, one at a time.	ALICE-LP — a new German short-answer dataset the authors built from real school formative assessments — plus the established English ASAP-SAS benchmark.	GRAASP and ToLeGRAA — attention-based transformer architectures for rubric-criterion alignment.	Both architectures were competitive with strong existing baselines on both datasets, while keeping the rubric-criterion alignment interpretable enough to show which criterion a given mark came from.	*ALICE-LP is a newly introduced dataset, so there is not yet an established body of comparative results against it, which makes it harder to judge how the reported gains would hold up on other rubric-scoring benchmarks.*

Research Gap Identification and Justification

Identified Research Gap

Across the base paper and the nine related papers, a consistent pattern shows up: rubric-grounded LLM scoring reaches strong agreement with human graders — Paper 1's ICC of 0.94, Paper 7's statistical equivalence to faculty scores — but that agreement is demonstrated on small, single-subject, single-institution datasets (30 scripts in Paper 1; a single question type in Paper 9), and it does not stay stable once scale, subject diversity, or answer difficulty increases. Paper 7's 1,450-response study shows agreement breaking down specifically for higher-order "Understanding" and "Evaluating" questions, and Paper 6's 42-language TIMSS deployment shows that at real scale, a meaningful share of AI-human disagreement is actually human-rater disagreement, not machine error. Meanwhile, only one paper in this survey (Paper 4) quantifies how much OCR error degrades a scoring pipeline's final accuracy, and no paper combines that OCR-robustness finding with explicit rubric grounding. There is therefore a clear gap for a system that stays reliable outside the lab conditions most of these papers are validated in — handwritten, multi-subject, OCR-dependent answer scripts of the kind an Indian engineering college actually produces, evaluated against more than one human rater rather than a single reference score.

Justification

- Dataset and scale bias: eight of the ten papers surveyed validate on a single subject, a single institution, or a single language, with sample sizes ranging from a few dozen scripts (Paper 1) to a few hundred (Paper 4); only Paper 6 (TIMSS, 42 languages) and Paper 3 (SAS-Bench, 9 subjects) test across meaningfully diverse conditions, and neither combines that scale with OCR-dependent handwritten input.
- OCR sensitivity is measured quantitatively in only one paper: Yadav et al. (Paper 4) show correlation with human graders falling from 0.89 to 0.81 as OCR error rate rises, but no rubric-grounded LLM paper in this survey (Papers 1, 7, 8, 9, 10) tests how their scoring approach holds up once the input text itself is noisy rather than clean.
- Burrows, Gurevych and Stein [12] documented, a decade before this survey, that automatic short-answer grading struggles with paraphrase, partial correctness, and answer-length variation — the same difficulty that Paper 9 addresses by decomposing answers into atomic scoring units and that Paper 10 addresses through explicit rubric-criterion alignment. That this problem is still being actively re-solved in 2025-2026 papers indicates it remains open rather than resolved.
- Human-rater agreement is itself a moving target: Zhang [13] found systematic differences between human and automated scoring behaviour even without any AI involved, and Paper 6 independently confirms this at TIMSS scale, showing that "accuracy against a human grader" is not always a fixed, trustworthy ground truth to benchmark against.

- Practical deployment constraint: no paper in this survey combines (a) OCR-error tolerance validated with real numbers, (b) explicit rubric-criterion grounding, and (c) formal inter-rater reliability against more than one human grader across more than one subject — leaving a gap for an integrated system that does all three together.
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Proposed Direction (basis for the future Mini Project)

A promising direction is to build a hybrid, rubric-guided Automatic Answer Sheet Evaluation system that pairs explicit rubric-criterion grounding (following the approach in Papers 1, 9, and 10) with an OCR-error detection and correction stage rather than passive tolerance of OCR noise (extending the finding in Paper 4), and to validate the resulting system against at least two independent human raters across more than one subject, rather than a single reference score — closing the reliability-validation gap that only Paper 1 and Paper 6 currently attempt.

SDG Mapping

Mapped Sustainable Development Goals

The research gap identified above, and the proposed mini-project direction (a hybrid, rubric-guided, OCR-error-tolerant answer sheet evaluation system validated against multiple human raters), align with the following United Nations Sustainable Development Goals:

SDG No.	SDG Title	Relevance to the Project
SDG 4	Quality Education	Consistent, rubric-guided automated grading reduces the time instructors spend on repetitive marking and frees that time for teaching, mentoring, and giving students more detailed, individual feedback — the core motivation stated across nearly every paper in this survey.
SDG 9	Industry, Innovation and Infrastructure	The range of techniques surveyed — OCR pipelines, retrieval-augmented LLM agents, cross-prompt pre-finetuning, attention-based rubric alignment — reflects active innovation in applying AI infrastructure to a concrete institutional assessment problem, which the proposed hybrid system extends further.

SDG No.	SDG Title	Relevance to the Project
SDG 10	Reduced Inequalities	A rubric-grounded, OCR-error-tolerant system applies the same marking criteria to every script regardless of handwriting quality, language, or which examiner happens to grade it, directly addressing the rater-inconsistency and cross-country grading disparities documented in Paper 6.
SDG 16	Peace, Justice and Strong Institutions	Transparent, rubric-traceable scoring (as in Paper 10's criterion-level alignment) makes it possible to show exactly which part of a rubric a mark came from, supporting fairer, more accountable and auditable academic assessment processes within an institution.

Justification of Mapping

- The primary motivation across the papers reviewed is reducing the grading burden on instructors while keeping assessment quality consistent, which links the work directly to SDG 4 (Quality Education).
- The proposed hybrid direction — combining rubric grounding, OCR-error tolerance, and multi-rater validation into one system — is itself an innovation-driven contribution to educational assessment infrastructure, aligning with SDG 9 (Industry, Innovation and Infrastructure).
- Because handwriting quality, language, and individual examiner bias currently produce inconsistent outcomes for different students (as shown in Paper 6's cross-country findings), a more robust and rater-independent system supports SDG 10 (Reduced Inequalities).
- Rubric-criterion-level transparency, as demonstrated in Paper 10, supports more accountable and auditable institutional assessment processes, connecting the work to SDG 16 (Peace, Justice and Strong Institutions).

Reference Papers

[1] H. M. T. W. Seneviratne and S. S. Manathunga, "Artificial intelligence assisted automated short answer question scoring tool shows high correlation with human examiner markings," BMC Medical Education, vol. 25, art. 1146, 2025.

[2] H. Funayama, Y. Matsubayashi, Y. Asazuma, T. Mizumoto, and K. Inui, "Cross-prompt Pre-finetuning of Language Models for Short Answer Scoring," International Journal of Artificial Intelligence in Education, vol. 35, pp. 2399–2420, 2025.

[3] P. Lai, K. Zhang, Y. Lin, L. Zhang, F. Ye, J. Yan, Y. Xu, C. He, Y. Wang, W. Zhang, and B. Cui, "SAS-Bench: A Fine-Grained Benchmark for Evaluating Short Answer Scoring with Large Language Models," arXiv:2505.07247, 2025.

[4] A. Yadav, Shalu, and S. Rathee, "AI-Driven Subjective Answer Evaluation: Integrating OCR and Semantic Similarity for Intelligent Learning Environments," Proceedings on Engineering Sciences, vol. 8, no. 1, pp. 311–320, 2026.

[5] M. G. Siddaraj, P. Sukesh, B. K. Nithin, M. S. Shreyas, and A. V. Hadapad, "Automated Evaluation of Handwritten Answer Script Using Deep Learning Approach," International Research Journal of Modernization in Engineering Technology and Science, vol. 6, no. 5, pp. 7038–7041, 2024.

[6] J. Y. Jung, L. Tyack, and M. von Davier, "Towards the implementation of automated scoring in international large-scale assessments: Scalability and quality control," Computers and Education: Artificial Intelligence, vol. 8, art. 100375, 2025.

[7] A. Rajan, S. M. Alexander, and C. L. Shenvi, "Can AI grade like a professor? Comparing artificial intelligence and faculty scoring of medical student short-answer clinical reasoning exams," Advances in Health Sciences Education, 2025.

[8] S. Haldar, S. Sengupta, and A. K. Das, "RUBRA: An Agentic AI System for Automatic Short Answer Grading Using LLMs and RAG," in Proc. 2025 Conf. on Intelligent Computing and Applications (CIACON), 2025.

[9] C. Zhang and L. Sun, "Fine-grained Rubric-Aware Automated Scoring of Programming Short Answer Questions via Large Language Models in Vocational Education," ACM, 2026.

[10] S. Gombert, Z. Sun, F. Zehner, J. Lossjew, T. Wyrwich, B. Czinczel, D. Bednorz, M. Kubsch, D. Di Mitri, K. Neumann, and H. Drachsler, "Are rubrics all you need? Towards rubric-based automatic short answer scoring via guided rubric-answer alignment," ACM, 2026.

[11] AD3501 Assignment Instructions Template, SRM TRP Engineering College, Dept. of Artificial Intelligence and Data Science, 2026.

[12] S. Burrows, I. Gurevych, and B. Stein, "The eras and trends of automatic short answer grading," International Journal of Artificial Intelligence in Education, vol. 25, no. 1, pp. 60–117, 2015.

[13] M. Zhang, "Contrasting automated and human scoring of essays," R&D Connections, no. 21, Educational Testing Service, 2013.

[14] M. Mohler and R. Mihalcea, "Text-to-text semantic similarity for automatic short answer grading," in Proc. 12th Conf. European Chapter Assoc. Comput. Linguistics (EACL 2009), Athens, Greece, 2009, pp. 567–575.